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EBU5608 Product Development and Management

Topic 18 – Start-ups and Entrepreneurship

1

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Agenda

- Product development in a small company
- What is a 'start-up'
- Entrepreneurship
- Advantages and Disadvantages of starting a new business
- Challenges of a small business
- Types of business
- Putting a high-tech start-up together
- Raising capital
- Business plan
- Causes of business failure

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2

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Introduction

- **Free enterprise** is the economic basis for all entrepreneurial activity
  - It means that any individual is free to transform an idea into a business
- The opportunities for potential entrepreneurs are unlimited
- During the last 10 years, new ventures have emerged at the rate of 500,000 per year in the US

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Product development in a small company

- We have talked about Product Development in **large companies**
- The same processes need to be used in **small companies** such as start-ups
- BUT:
  - There will be **less people** involved, and they may each have **multiple** roles
  - You may need to **simplify** some of the processes
- Do **not** make the mistake of thinking "we are too small to bother with a proper process"
- You need to understand the special **challenges** of starting a company to exploit your ideas

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What is a 'startup' [2]

- A **startup company** or **start-up** is a company with a limited operating history
- These companies, generally newly created, are in a phase of development and research for markets
- Typically, it involves some **novel technology, or use of technology**, but crucially…
  - a coherent means for **making money** from the technology
- High-tech (e.g. telecom, software) is very different from bio-tech or medical
- Startup x spin-out

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The entrepreneur [3]

- An **entrepreneur** is a person who
  - has possession over a new enterprise or venture and
  - assumes full accountability for the inherent risks and the outcome
- **Entrepreneurship** is the practice of starting **new** organisations or **revitalizing** mature organisations,
  - particularly new businesses - generally in response to identified opportunities

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6

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Two types of entrepreneurship [1]

Social

- Aim is to create social change rather than make money
- Often involves business, public & charity organisations
- Examples
  - poverty relief
  - environment
  - arts & culture

Technological

- Aim is to gain independence to exploit a technology
- Often results from being frustrated in a large company

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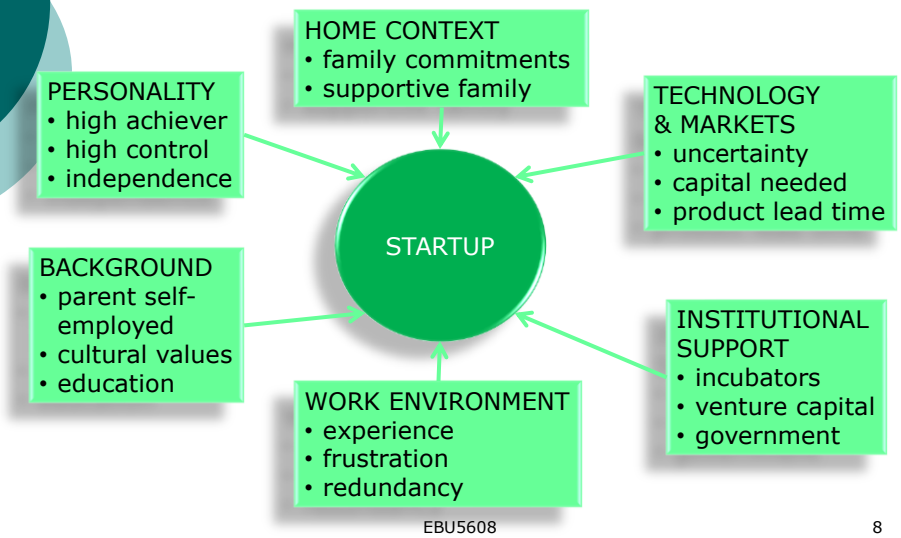
Factors influencing the successful creation of startups



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8

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Supporting entrepreneurship [4]

- Many kinds of organizations now exist to **support** would-be entrepreneurs, including
  - specialized **government agencies**
    - e.g. in the UK, Business Link is a free business advice and support service, available online and through local advisers [5]
  - **business incubators**
  - **science parks, and**
  - **Non-Government Organisations**
    - e.g. Chambers of Commerce in the UK [6]

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Primary challenges

- **Limited Resources**:
  Startups often face financial and human resource constraints, hindering R&D, hiring talent, and scaling the product, which can delay development and innovation.
- **Market Competition**:
  The tech sector is highly competitive, with established companies dominating the market. Startups must innovate rapidly and differentiate themselves to attract customers and investors.
- **Customer Adoption and Trust**:
  Convincing customers to adopt a new product is challenging, especially with concerns about privacy, security, and compatibility. Customer education and trust-building are essential for success.

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Primary challenges

- **Cash Flow Management**:
  Startups often struggle with managing cash flow, especially during the early stages. They may face difficulties in balancing the inflow and outflow of funds, which can delay product development, marketing efforts, and operational activities.
- **Talent Acquisition and Retention**:
  In the tech sector, finding and retaining skilled employees can be difficult. The demand for talent, especially in specialized fields such as software engineering, AI, and cybersecurity, can lead to intense competition for the best candidates.
- **Scaling Challenges**:
  As the company grows, managing the scaling process efficiently becomes a challenge. Issues such as scaling the infrastructure, maintaining product quality during rapid growth, and expanding the team while keeping culture intact are common hurdles.

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11

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Primary challenges

- **Intellectual Property (IP) Protection**:
  Startups in the tech sector often rely heavily on unique innovations. Protecting their intellectual property (such as patents, trademarks, and copyrights) from competitors can be a critical challenge, especially when resources are limited for legal protection.
- **Funding and Investment**:
  While funding is essential for growth, it can also be difficult for tech startups to secure investments. Venture capitalists and angel investors often require substantial evidence of market fit and scalability, and competition for funding is high.

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Primary challenges

- **Regulatory Compliance**:
  The tech industry is subject to increasing government regulation, especially with data privacy laws (e.g., GDPR, CCPA). Navigating legal compliance, especially around data protection and security, can be overwhelming for startups.
- **Customer Feedback and Iteration**:
  Early-stage tech companies must quickly adapt their products based on customer feedback. Building a feedback loop for **product iteration** is crucial, but this requires both **resources** and **time**, which may be scarce in a startup.

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Pros and cons of starting a new business

Advantages

- Independence
- Financial opportunities
- Community service
- Job security
- Family employment
- Challenge

Disadvantages

- Sales fluctuations
- Competition
- Increased responsibilities
- Financial losses
- Employee relations
- Laws and regulations
- Risk of failure

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14

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Launching a Smart Fitness Device Startup

Let us consider a scenario of starting a **smart fitness device** company, which provides wearables that track and improve users' health and fitness through advanced sensors, data analytics, and app integration.

You need to take some early decisions about your company:

- What **type of business** – lifestyle?
- What form of **ownership** – sole trader?
- Are you driven by the **technology or** the **market**?
- Who are your **customers**?
- How will you **raise capital**?

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What type of business? (Lifestyle vs. Scalability)

- **Lifestyle Business**: Your goal would be to create a product that allows you to maintain a balance between work and personal life, with steady revenue streams, but not necessarily rapid growth. The product would be a niche item, focusing on a small community of fitness enthusiasts who value personalized health insights.
- **Scalability Focused Business**: If you decide to scale, the aim would be to disrupt the fitness industry by creating a device with advanced features (like AI-powered health advice, integration with fitness apps, etc.) that can appeal to a broader market, both domestically and internationally.

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What form of ownership? (Sole Trader, Partnership, or Corporation)

- **Sole Trader**: You take full responsibility for the company, and it is easier to manage with fewer administrative tasks. However, the downside is the personal liability in case of financial trouble.
- **Partnership**: If you and your co-founder have complementary skills and are willing to share responsibilities, a **partnership** may be a good choice. This allows shared resources, but you'll also share the risks and profits.
- **Corporation (Ltd)**: As your business grows and you seek outside investors, you might consider setting up as a **limited company** or **corporation**. This protects your personal assets, but it involves more regulatory requirements, such as tax filings and official accounting, and it's harder to change ownership.

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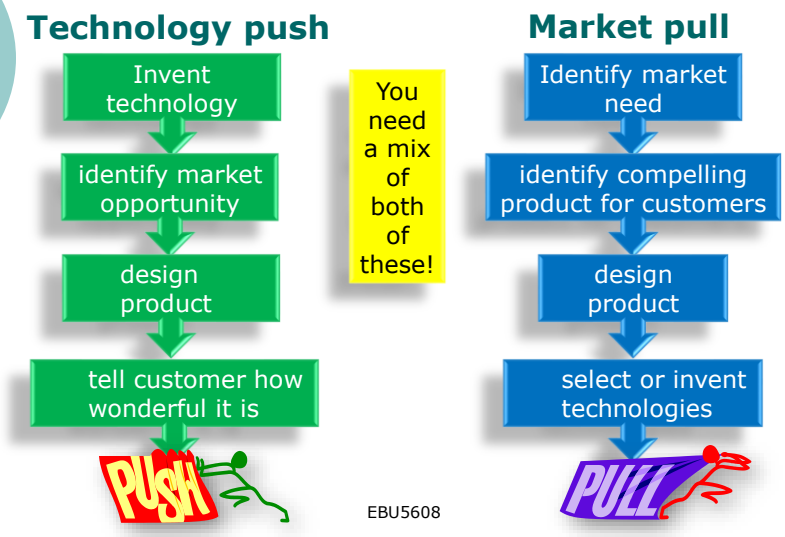
Are you driven by the technology or the market?



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Who are your customers?

- End-users, i.e. consumers
  - i.e. you are B2C (business to consumer)
- Business that has its own customers
  - i.e. you are B2B (business to business)
- You **must** know who your potential customers are

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How will you raise capital?

- **Self-Funding (Bootstrapping)**: You and your co-founders could choose to fund the startup from your own savings. This gives you full control over the company but comes with the risk of losing personal assets if the business fails.
- **Angel Investors**: You may seek **angel investors** who are individuals willing to invest in early-stage businesses. Angel investors can offer not only funding but also valuable **mentorship** and **connections**.
- **Venture Capital (VC)**: If you aim for rapid growth, you may decide to pursue **venture capital funding**. VC firms can provide large sums of money, but they will usually require equity in return and want a say in the business strategy.
- **Crowdfunding**: Another option is to launch a **crowdfunding campaign** through platforms like **Kickstarter** or **Indiegogo**, where you can raise small amounts of money from a large number of people who are interested in your product.

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Fund-raising Stages

- Pre-investment
  - University fund, friends, bank loan
  - £5k-£50k
  - build demonstrator, build business plan

- Round A
  - Angels, or special startup funds
  - £100k-£500k
  - Establish board, partial executive team
  - 5 - 10 employees

- Round B
  - Venture Capital
  - £1M-£5M
  - Complete board and exec team. Expand to 20+

- Exit

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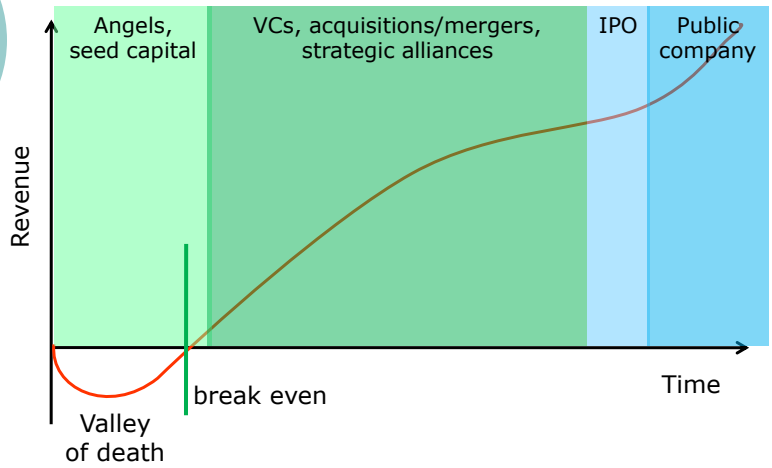
Start-up financing cycle



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How a hi-tech start-up comes together

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The ingredients:

- Technology
- Intellectual Property (Protection)
- The Vision (Where can the technology lead? Is there a market for it)
- Personnel
  - Scientific leaders
  - Business managers
  - Fund Raisers
  - Scientists/Engineers
- Money

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The stages for creating a hi-tech start-up

1. **Assess** the opportunities - generate, evaluate & refine the business concept
2. **Develop** the business plan and decide on the structure of the venture
3. **Acquire** the necessary resources and funding
4. **Grow** and harvest the venture

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Business plan

- Business plan has two fundamental functions:

1. Guiding the company by charting its future course and defining its strategy for following it.
2. Attracting lenders and investors who will provide needed capital.

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Business Plan

- Is the entrepreneur's **road map** for a successful enterprise
- Describes
  - The **market** you plan to enter
  - The unique and compelling **features** of your contribution to this market
  - **Intellectual property**
  - **Business model** and **Financials**
  - Team and Company development and **strategy**
  - **Investment** proposition
  - **Exit** strategy

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Features vs. Benefits

- **Feature** – a descriptive fact about a product or service:
  "an ergonomically designed, more comfortable handle"
- **Benefit** – what a customer gains or a satisfaction received from the product or service feature:
  'better fuel consumption by a new car engine which reduces overall costs and increases efficiencies'

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Why do you need a Business Plan?

A business plan…

- provides a **formal agreement** between the founders of a company about the direction to be taken
- can reduce **self-delusion** amongst the founders
- defines **responsibilities** & rewards
- helps to translate **abstract** goals into **explicit** operational needs

Without a business plan…

- nobody will invest in your company
- you will wander aimlessly

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Business Model

- The Business Plan must describe your Business Model
- This shows how you will **make money** from your business, e.g.
  - Selling software
  - Selling services
  - Licensing software
  - Selling hardware
  - Selling/licensing Intellectual property
    - e.g. the ARM processor (CPU based)
  - Some mix of the above
- Will be supported by **Revenue predictions**

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Key Elements of a Business Plan

- Title Page and Table of Contents
- Executive Summary
- Vision and Mission Statement
- Company History
- Business and Industry Profile
- Business Strategy
- Description of Products/Services

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Key Elements of a Business Plan

- Marketing Strategy
  - Document market claims
  - Show customer interest
- Competitor Analysis
- Description of Management Team
- Plan of Operation
- Projected Financial Statements
- Loan or Investment Proposal

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Exit strategy

- VC wants to know how it will get a return on investment (ROI)
  - Company **sale** (e.g. to Microsoft, Google, Philips, etc)
  - **IPO** (Initial Public Offering), i.e. making shares publicly available via a stock exchange (SSE – Shanghai Stock Exchange; AIM (London) – Alternative Investment Market, NASDAQ – New York, etc.)
- In how many **years**?
- If you get this right, you are now getting 10% of the value of a company worth £50M or more!

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Guidelines for Preparing a Business Plan

- Remember: No one can create your plan *for* you.
- Potential lenders want to see financial projections, but they are more interested in the *strategies* for reaching those projections.
- Show how you plan to set your business apart from competitors; don't fall into the "me too" trap (copying other businesses – no advantage to this).
- Identify your target market and offer evidence that customers for your product or service exist.

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Causes of business failure

Business failure is caused by a number of reasons but the most common of these are detailed below:

- lack of **skills**
- **sales** problem
- **financial** control
- lack of **funds**
- high cost of **finance**
- insolvent **customers**
- overtrading **development**
- **marketing** issues
- red tape (**bureaucracy**)

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Summary

- There are no guarantees for success.
- Creating a business plan will be valuable primarily because of the process itself.
- The business planning process may provide insight to increase the chances for success.
- The business plan: Entrepreneurs benefit; lenders and investors demand it!

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35

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38

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EBU5608 39

39